

- 7 A square coil of side 5.0 cm and 50 turns is placed horizontally midway between the poles of a magnet as shown in Fig. 7.1.

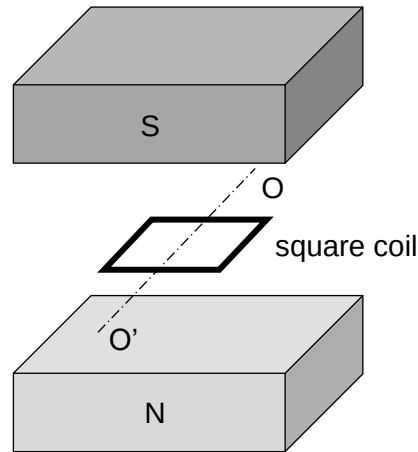


Fig. 7.1

The magnetic flux density due to the magnet in the area of the coil may be regarded as uniform and acts in a vertical direction with a magnitude of 0.12 T.

- (a) Calculate the magnetic flux through the area of the coil.

$$\text{magnetic flux} = B A = 0.12 \times 0.50^2 = 3.0 \times 10^{-4} \text{ Wb} \quad [\text{A1}]$$

magnetic flux = Wb [1]

- (b) The coil can be displaced by any angle θ about its axis OO' as shown in Fig. 7.2.

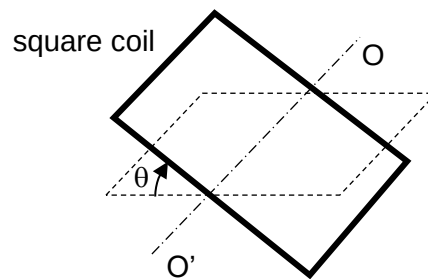


Fig. 7.2

Calculate the magnetic flux linkage in the coil when $\theta = 30^\circ$.

$$\text{magnetic flux linkage} = N B A \cos 30^\circ = 50 \times 3.0 \times 10^{-4} \cos 30^\circ \quad [\text{M1}]$$

$$= 0.013 \text{ Wb} \quad [\text{A1}]$$

magnetic flux linkage = Wb-turns [2]

- (c) The coil is rotated about OO' with a constant angular frequency. With reference to Faraday's law of electromagnetic induction,

- (i) explain why a current is present in the coil,

As the coil rotates, the angle between the magnetic field and the plane of the coil changes, leading to a change in the magnetic flux linkage in the coil. [B1] By Faraday's law, an e.m.f. is induced in the coil which will produce a current in the closed loop. [B1]
.....[2]

- (ii) state and explain the value(s) of θ within one rotation of the coil at which the current in the coil is the greatest.

When $\theta = 90^\circ$ and 270° , [A1] the rate of the change in the flux linkage is the greatest [M1], thus by Faraday's Law, the e.m.f. induced at that instant is also the greatest, resulting in the greatest current.
.....[2]

- (d) Fig. 7.3 shows the current induced in the coil when it is rotating with a period of 50 ms.

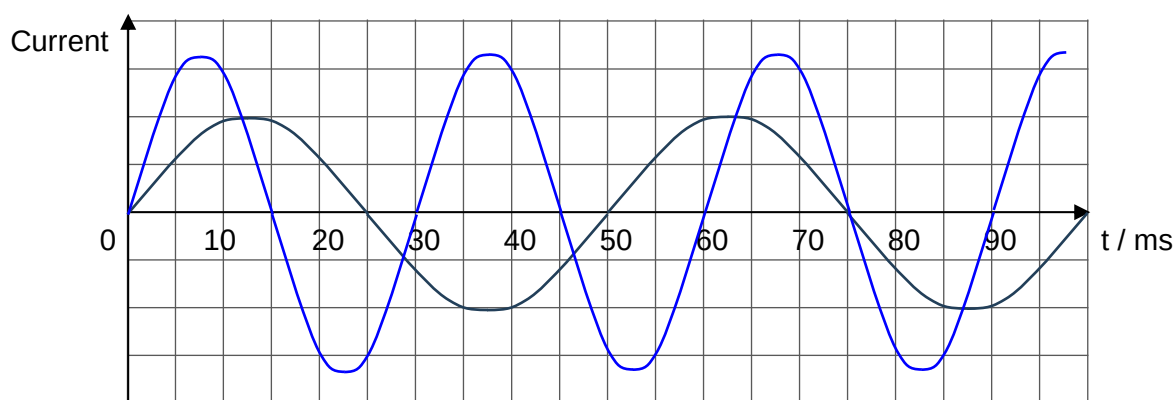


Fig. 7.3

- (i) Without further calculations, draw another graph on Fig. 7.3 to show the current in the coil when the period of rotation is decreased to 30 ms. [1]

- (ii) Determine the ratio of

$$\frac{\text{mean power required to rotate the coil with a period of 30 ms}}{\text{mean power required to rotate the coil with a period of 50 ms}}$$

$$P = \frac{E^2}{R} \text{ and } E = -\frac{d\Phi}{dt} \propto \omega \propto \frac{1}{T} \rightarrow P \propto \frac{1}{T^2} \text{ [M1]}$$

$$\text{ratio} = \left(\frac{50}{30}\right)^2 = 2.8 \text{ [A1]}$$

$$\text{ratio} = \dots\dots\dots [2]$$

[Total: 10]

8 Recent developments show that there has been a significant surge in the number of commercial